



Bangladesh University of Engineering and Technology

ME 352: Machine Design Sessional

Report On

Stress Analysis of Key Components in a Crane Pulley System

Group No: 5 (B2)

Author: 2010119 - Kazi Md. Zareer
2010118 - Md. Abu Ozayer Rifad
2010104 - Md. Amanullah Asif
2010093 - Ibnul Muhtadi Shah
2010091 - Khalilur Rahman

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1 Introduction

Stress analysis is the core of mechanical design, useful for studying the response of structures and parts under load. This paper therefore discusses stress analysis of the crane pulley system because the crane is used primarily for material handling and lifting operations. Crane pulley systems consist of several different parts working together— pulley, shaft that supports it, bearings, and base structure. Since each part is under different stress because of applied loads, the structural integrity of each individual part must be analyzed.

The main function of the pulley system is to lift and move large loads by transmitting force through ropes or cables. At work, the pulley system components experience different types of forces, including tension in the rope, bending moments on the shaft, and reaction forces on the bearings and base. Any unreasonable design or unexpected loading conditions may result in failure of the component, hence compromising the safety and performance of the crane. Our objective is to evaluate:

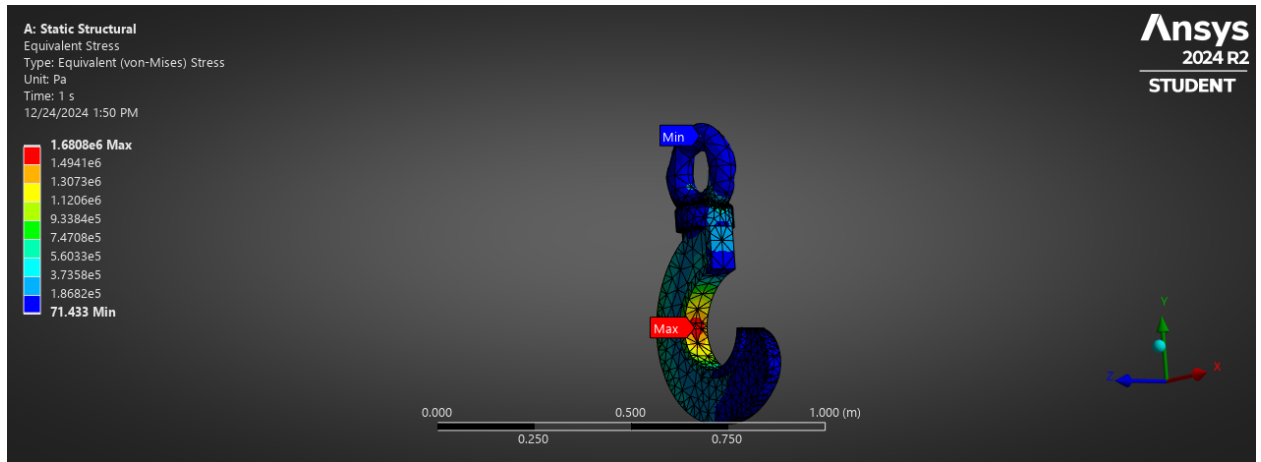
1. The stress experienced by the crane hook, supplemented with corresponding hand calculations.
2. The stress on the V-groove pulley, which facilitates lifting the load using a rope.
3. The stress on the cantilever-shaped beam of the crane, critical for supporting the overall structure.



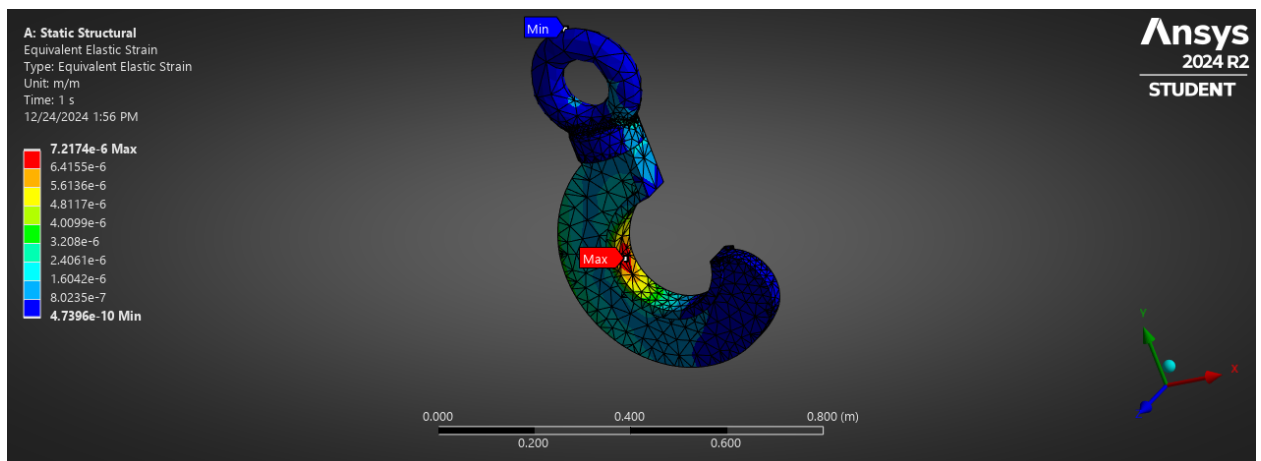
Figure 1: A Simple Crane Pulley System

2 Ansys Analysis

2.1 Stress Analysis of Crane Hook

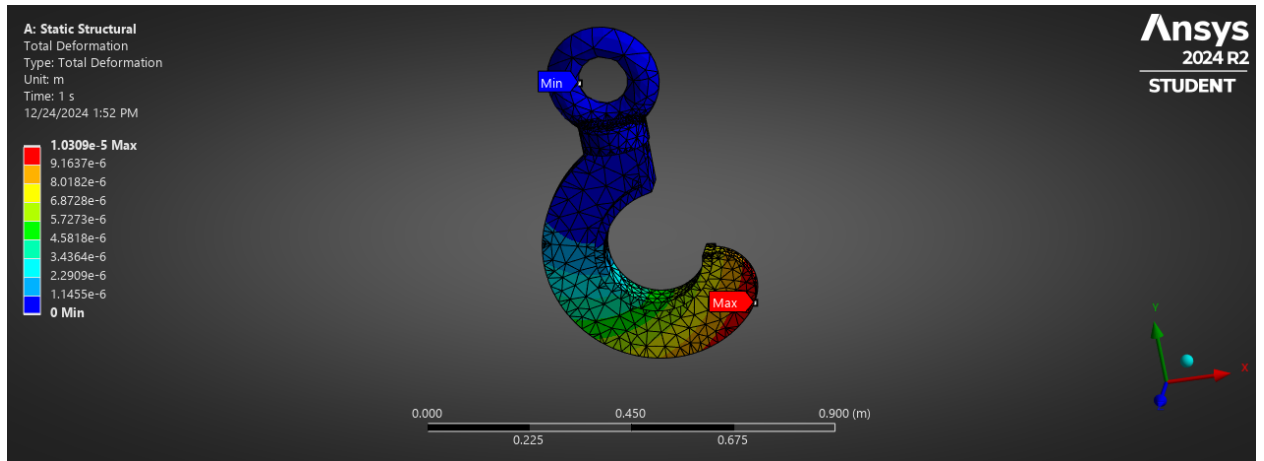


(a) Equivalent (Von-Mises) Stress.

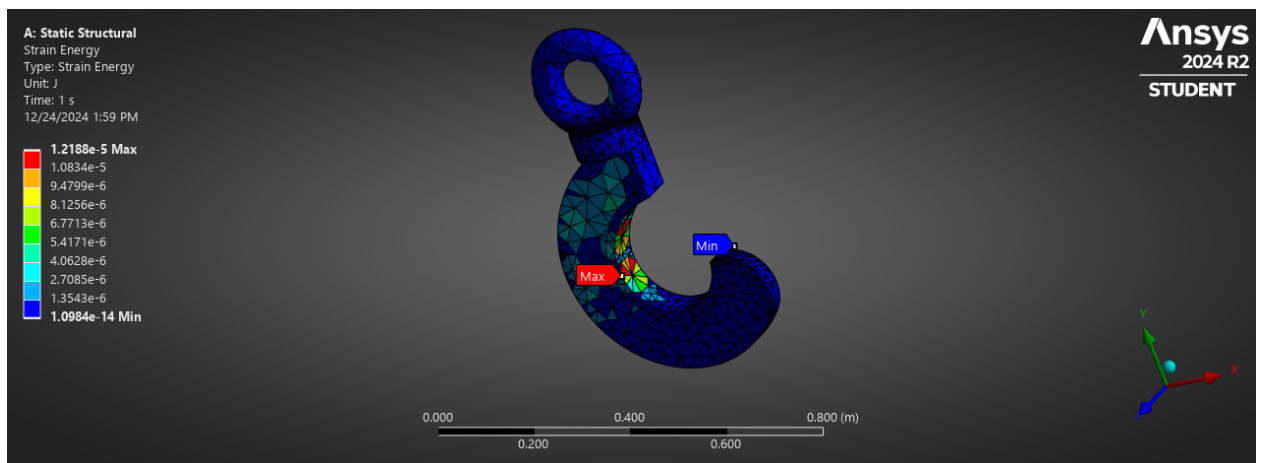


(b) Equivalent (Von-Mises) Strain.

Figure 2: Ansys analysis results: Stress and Strain.



(a) Total Deformation.



(b) Strain Energy.

Figure 3: Ansys analysis results: Deformation and Strain Energy.

2.2 Stress analysis of V-groove Pulley

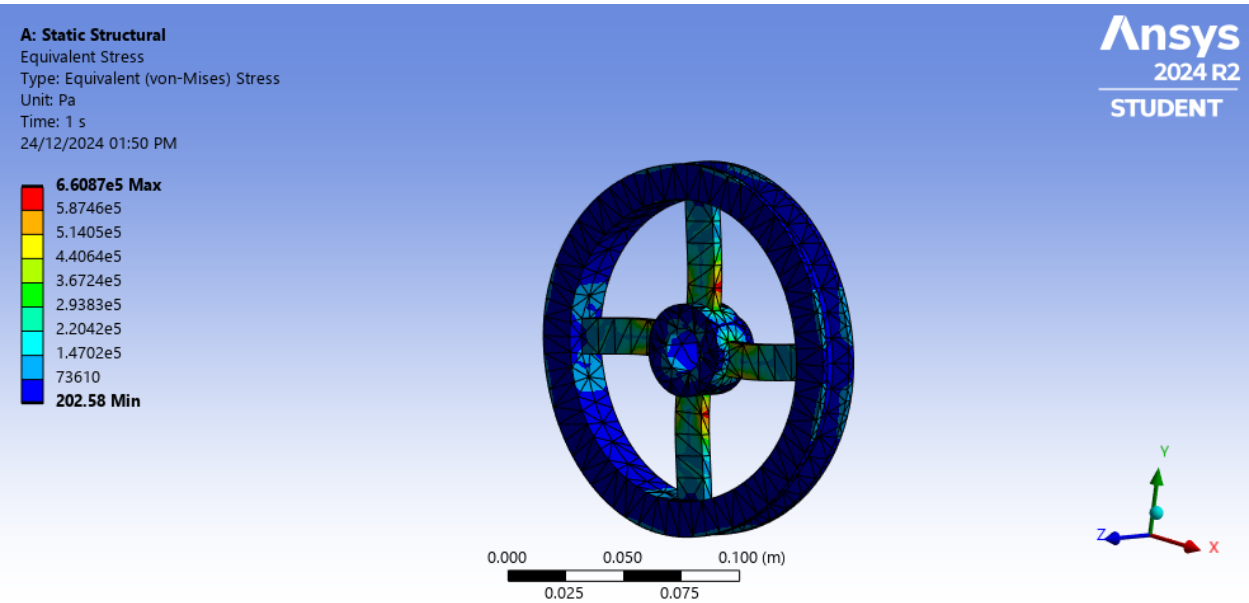


Figure 4: Equivalent (Von-mises) Stress on V-groove pulley.

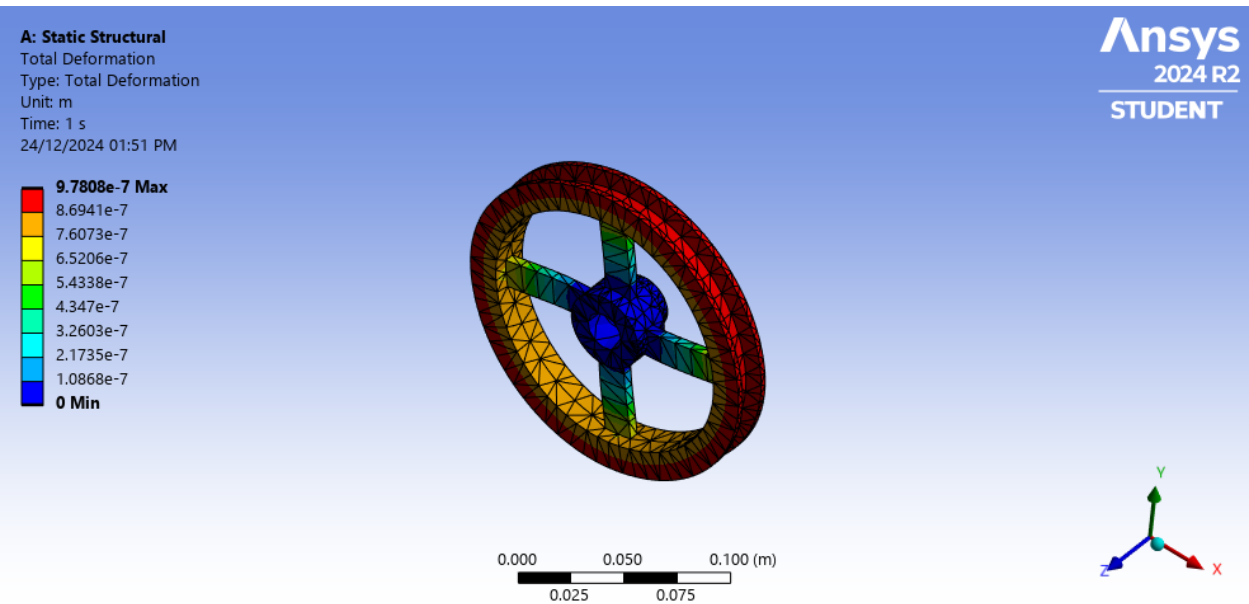
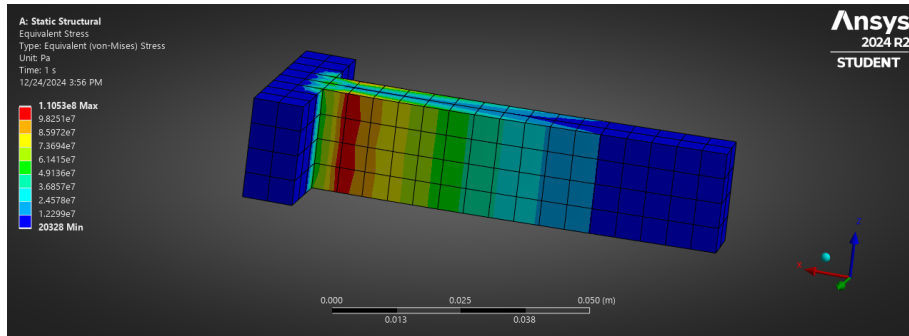
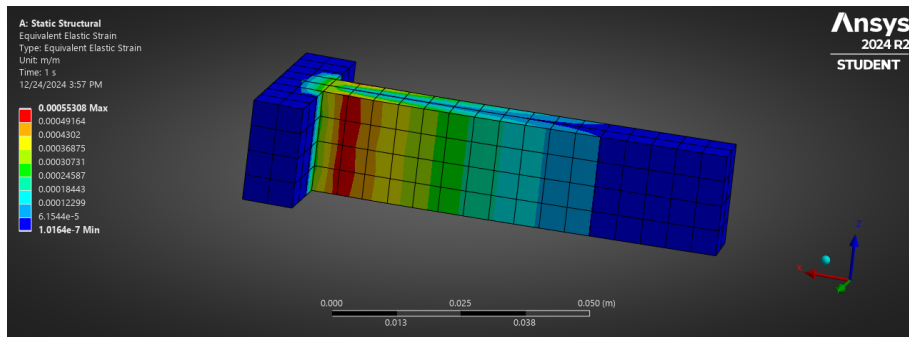


Figure 5: Total Deformation.

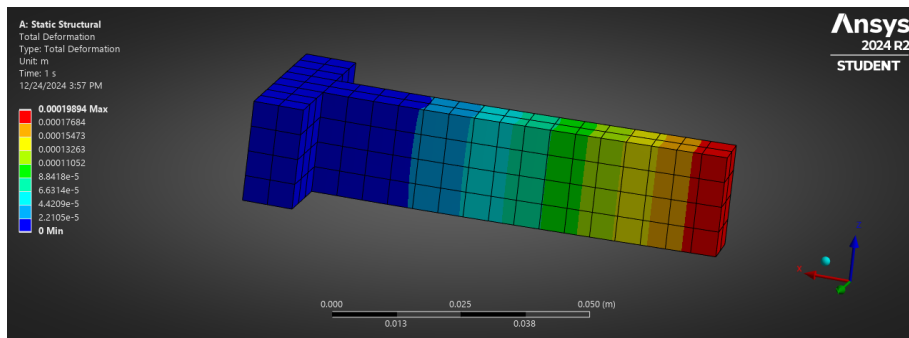
2.3 Stress analysis of Cantilever-shaped beam of the crane



(a) Equivalent (Von-mises) Stress generates at the fixed support of the crane beam.



(b) Strain in the beam



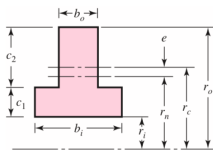
(c) Total Deformation in the beam.

Figure 6: Stress and Strain analysis of the fixed cantilever shaped beam

3 Stress analysis of Crane Hook (Hand Calculation)

Here is the hand calculation of the stresses on the crane hook:

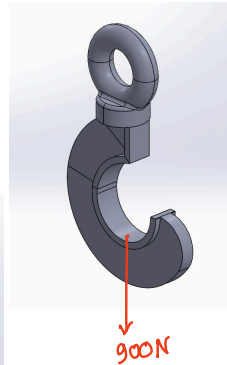
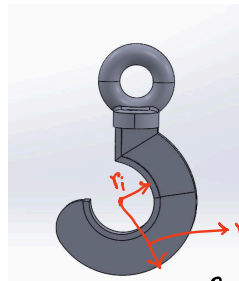
Calculation:



$$r_c = r_i + \frac{b_i c_1^2 + 2b_o c_1 c_2 + b_o c_2^2}{2(b_o c_2 + b_i c_1)}$$

$$r_n = \frac{b_i c_1 + b_o c_2}{b_i \ln[(r_i + c_1)/r_i] + b_o \ln[r_o/(r_i + c_1)]}$$

Here, $r_i = 106.6 \text{ mm}$
 $c_1 = 20 \text{ mm}$
 $c_2 = 117.26 \text{ mm}$
 $b_i = 75 \text{ mm}$
 $b_o = 40 \text{ mm}$



$B = r_i + c_1 + c_2 = 243.86 \text{ mm}$

Now, $r_c = r_i + \frac{b_i c_1^2 + 2b_o c_1 c_2 + b_o c_2^2}{2(b_o c_2 + b_i c_1)}$

$$= 106.6 + \frac{75 \times 20^2 + 2 \times 40 \times 20 \times 117.26 + 40 \times 117.26^2}{2(40 \times 117.26 + 75 \times 20)}$$

$$= 168.6 \text{ mm}$$

$r_n = \frac{b_i c_1 + b_o c_2}{b_i \ln\left(\frac{r_i + c_1}{r_i}\right) + b_o \ln\left(\frac{r_o}{r_i + c_1}\right)}$

$$= \frac{75 \times 20 + 40 \times 117.26}{75 \ln\left(\frac{106.6 + 20}{106.6}\right) + 40 \ln\left(\frac{243.86}{106.6 + 20}\right)}$$

$$= 158.246 \text{ mm}$$

Now, $e = r_c - r_n = 168.6 - 158.246 = 10.35 \text{ mm}$
 $c_i = r_n - r_i = 158.246 - 106.6 = 51.646 \text{ mm}$
 $c_o = r_o - r_n = 243.86 - 158.246 = 85.614 \text{ mm}$

$$\therefore M = F \times l_2 = 900 \times 168.6 \text{ Nmm}$$

$$= 151.74 \times 10^3 \text{ Nmm}$$

$$A = b_1 c_1 + b_2 c_2 = 75 \times 20 + 40 \times 17.26 = 6190.4 \text{ mm}^2$$

stress at inner surface,

$$\sigma_i = \frac{M c_i}{A e r_i} = \frac{151.74 \times 10^3 \times 51.646}{6190.4 \times 10.35 \times 106.6}$$

$$= 1.147 \text{ MPa} \quad \text{inner surface}$$

Stress at outer surface,

$$\sigma_o = \frac{M c_o}{A e r_o} = \frac{151.74 \times 10^3 \times 85.614}{6190.4 \times 10.35 \times 243.86}$$

$$= 0.83 \text{ MPa} \quad \text{outer surface}$$

4 Conclusion

The stress analysis of the crane pulley system was conducted using a combination of ANSYS simulations and hand calculations, providing insights into the structural performance of its key components.

For the crane hook, finite element analysis (FEA) in ANSYS revealed the stress distribution under various loading conditions. Hand calculations were performed to validate the simulation results. Areas of high stress and potential failure were identified, which would facilitate targeted design improvements to enhance load-carrying capacity.

In the V-groove pulley, the analysis focused on stress induced by rope tension during lifting operations. We assumed 1 Nm of moment induced by the rope. ANSYS simulations pinpointed stress concentrations at the groove and contact points, leading to recommendations for material selection and structural enhancements to improve reliability.

The cantilever-shaped crane beam was analyzed for bending and shear stresses. FEA simulations highlighted critical stress regions. Deformation contours and animations helped predict the beam's behaviour under load.

This integrated approach combined advanced simulation tools and traditional calculations to provide a good understanding of the mechanical behavior of the crane system, as well as an overview of the FEA simulation process.